# Adaptive Quantum Error Mitigation Under Fixed Shot Budgets: Operating Regimes, Abstention, and Noise-Model Shift

Zhen-Jun Liu

November 6, 2026

## Abstract

Quantum error mitigation (QEM) trades bias reduction against auxiliary executions and sampling variance, so a method that helps one circuit may waste resources or become unstable elsewhere. We compare unmitigated estimation, full-assignment-matrix readout mitigation, and linear zero-noise extrapolation (ZNE) under identical total shot budgets. The simulator benchmark combines GHZ and fixed-seed hardware-efficient circuits with widths 2, 4, and 6, depths 1, 3, and 6 where applicable, four main noise levels, three shot budgets, and 50 deterministic repeats. A cost-aware oracle labels 144 operating cells as 77 raw winners, 26 practical ties, 18 readout winners, 14 ZNE winners, and 9 inconclusive, after correcting an implementation error that had labeled every decisive raw win as a tie. A frozen, non-ML rule selector reaches 41.4% decision compatibility on a 29-cell deterministic holdout and 50.0% under four synthetic noise shifts. A consequence audit shows that fixed policies match or beat it on several measures even before its diagnostic cost is charged. A preregistered runtime selector with charged pilot shots then fails all three frozen criteria on 120 cells of unseen circuits, a no-go result. The benchmark retains 99 explicit readout failures and 179 of 432 method summaries with negative RMSE gain. An additional preregistered ZNE validity gate accepts only 5 of 16 simulated circuit–noise–budget cells; none of the 8 tested hardware-efficient cells passes every check. In a separate factorial study, combined readout mitigation and ZNE has the lowest expectation RMSE in 12 of 16 preregistered conditions, yet all 16 combined-ZNE conditions fail at least one frozen validity criterion. Its near-tie decision-error probability remains between 0.26 and 0.48. In a fixed-budget MaxCut application, lower energy RMSE does not reliably improve top-1 QAOA-candidate selection, and only 2 of 828 completed selections certify a winner under simultaneous intervals. Sequential stopping with corrected repeated looks saves shots on clear decisions while keeping false certification below 3% at zero margin, compared with 22–25% for uncorrected peeking. It cannot detect bias: under strong gate-noise imbalance, raw estimation certifies the wrong candidate in every repeat after about 2,000 shots. Under synthetic readout drift, triggered recalibration uses about 40% fewer shots than recalibrating at every decision with similar reliability, whereas indefinite reuse fails under directed drift. A hardware readiness review of six candidate cells ends in a documented no-go, with no job submitted: only 1 of 7 go requirements is met. These simulated results support selective mitigation and abstention as research questions, but do not validate a general selector or establish hardware performance.

## 1 Introduction

QEM can reduce expectation-value bias without logical encoding, but its additional circuits and statistical amplification consume finite resources [10, 6, 3]. Comparing methods at unequal execution cost can therefore manufacture an apparent advantage. Even under a fixed budget, the best method can change with circuit structure, observable, noise, and sampling regime.

This work asks when mitigation improves RMSE at fixed total shots, when methods are practically indistinguishable, whether a simple selector can choose safely without the exact reference, when apparent ZNE improvement is credible, and whether readout correction and ZNE cooperate under one resource ledger. We also ask whether a mitigated energy estimate improves a complete MaxCut/QAOA candidate-selection decision rather than only an expectation metric. Finally, we ask whether a decision can stop early without losing error control, how long a calibration can be reused, and what the selector's outcomes cost rather than only how often its labels match. Last, we ask whether running on a device could currently discriminate between explanations. The contribution is a reproducible simulator benchmark with repeat-level data, explicit execution ledgers, cost-aware labels, abstention rules, and evaluation under noise-model mismatch. It is not a proposal for new QEM theory or a claim of quantum-hardware advantage.

## 2 Methodology

### 2.1 Circuit families and observables

The GHZ family prepares

$$|\mathrm{GHZ}_w\rangle = (|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}, \tag{1}$$

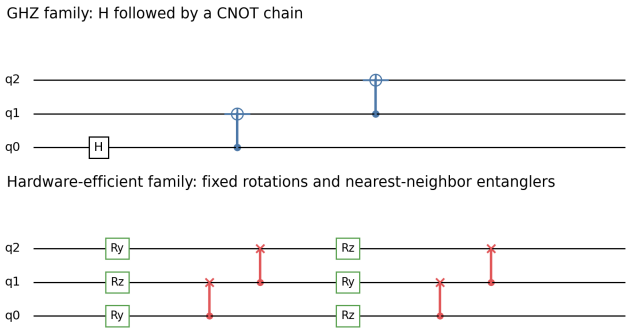



Figure 1: The benchmark combines an analytically controlled GHZ family with a nontrivial hardware-efficient depth grid; both diagrams are conceptual and all executed circuits are generated deterministically in code.

with analytic observables $X^{\otimes w} = 1$ and neighboring $ZZ = 1$. The hardware-efficient family alternates fixed-seed single-qubit rotations and nearest-neighbor entanglers at depths 1, 3, and 6. Its Pauli observables and statevector references are fixed before noisy sampling; some references are deliberately away from $\pm 1$ to avoid a stabilizer-only benchmark.

For repeat $r$, $e_r = \widehat{\mu}_r - \mu_{\text{ref}}$. We report bias, sample variance, and

$$\text{RMSE} = \sqrt{R^{-1} \sum_{r=1}^{R} e_r^2}. \tag{2}$$

Paired RMSE-gain intervals use deterministic bias-corrected and accelerated bootstrap resampling. Decision-error probability is the fraction of repeats whose estimated sign disagrees with the reference decision; Wilson intervals quantify its binomial uncertainty.

## 2.2 Mitigation and fixed budgets

Raw estimation spends the full budget on the target. Readout mitigation allocates 20% to all $2^w$ calibration states and 80% to the target. It solves a constrained assignment-matrix correction and reports ill-conditioning explicitly rather than silently inverting an unstable matrix [1]. Linear ZNE splits the same budget over global folds $\lambda \in \{1, 3, 5\}$ and uses the preregistered fit

$$\widehat{\mu}(\lambda) = \beta_0 + \beta_1 \lambda, \qquad \widehat{\mu}_{\text{ZNE}} = \beta_0, \tag{3}$$

without clipping out-of-range extrapolates [4].

## 2.3 Combined mitigation and factorial interaction

The combined estimator calibrates one full assignment matrix per repeat, corrects the probability distribution at each fold, converts each corrected distribution to an expectation, and then applies the same linear extrapolation. It reserves 20% of the total budget for calibration. The remaining shots are either divided equally across folds (balanced) or allocated 50%/30%/20% to $\lambda = 1, 3, 5$ (base-heavy). Intermediate counts, corrected estimates, extrapolates, and every calibration and folded execution remain in the repeat-level ledger. For paired absolute errors $E$, the factorial interaction is

$$I = E_{\text{combined}} - E_{\text{readout}} - E_{\text{ZNE}} + E_{\text{raw}}. \tag{4}$$

Negative $I$ indicates that the joint error is below the additive single-method contrast; it does not by itself establish a credible extrapolation mechanism.

## 2.4 ZNE validity and falsification

An extrapolate is not treated as credible merely because it moves toward the exact reference. A preregistered gate additionally requires retained signal at the largest fold, physical expectation and signed-probability diagnostics, positive paired gain with uncertainty, a positive advantage over matched-cost garbage folding, and no excess of destroyed or unstable repeats. Garbage folding increases execution cost without implementing the intended unitary-equivalent noise scaling and therefore serves only as a negative control. The gate uses 50 deterministic repeats per cell; rejected cells retain every failed criterion and are never converted into ZNE wins.

## 2.5 Cost-aware labels and frozen selector

For each fully completed cell, the oracle combines RMSE with an execution-overhead penalty. Differences below the preregistered RMSE threshold 0.005 are labeled `practical_tie`; incomplete three-method comparisons are `inconclusive`. A deterministic hash split reserves at least 20% of cells. The selector uses width, depth, declared noise regime, shots, calibration conditioning, and ZNE fit residuals. It can select raw, readout, or ZNE, or abstain; it never receives the statevector reference or oracle label.

Two limits apply to this baseline. Its diagnostics are 50-repeat cell means, which a runtime policy would not have, and acquiring them is not charged to the selector. Its rules also name the shifted noise channels, so the shift test measures known channel types rather than unknown ones.

The first implementation of the oracle could not assign the label `raw`: a raw winner has zero gain over itself, so it always reached the practical-tie branch. This paper reports corrected labels, in which a raw winner with a utility gap of at least 0.005 is labeled `raw`; every other rule is unchanged. Corrected regret is selected RMSE minus the lowest RMSE in the cell, so it cannot be negative, and abstaining cells are counted separately. Because the original results had already been inspected, this reanalysis is diagnostic rather than an independent holdout. The original labels remain in the repository.

## 2.6 Consequence audit and reliability ranges

The audit assigns each selection to one category. A compatible selection is useful or tied. An incompatible selection has negligible regret (at most 0.005), material regret, or catastrophic regret (at least 0.05 or at least $|\mu_{\text{ref}}|$). An abstention avoided failure if the selector with its abstention gates removed would have chosen a method with incomplete repeats or regret above 0.005; otherwise it was unnecessary. The same rules score always-raw, always-readout, and always-ZNE policies.

A method is reliable in a cell if it completes all 50 repeats with regret at most 0.005. A feature level is supported if it has at least 10 rule-design cells, a reliable rate of at least 80%, and a Wilson lower bound of at least 50%. Stability is the share of two-stage bootstrap resamples, over repeats and then cells, in which that level stays supported.

## 2.7 Preregistered runtime selector

A second selector sees only one repeat's pilot, which spends 25% of the budget on assignment calibration and on target samples at folds 1 and 5. It never receives a reference, a noise name, or another repeat. It returns `request_more_shots` if the fold-1 estimate is within two standard errors of zero; `recalibrate` if the calibration is singular or its condition number is at least 8; `invalid_zne_scaling` if folding significantly reverses or grows the signal; `abstain` for $|\widehat{\mu}_1| \geq 0.5$ with condition number at least 3; readout for $|\widehat{\mu}_1| \geq 0.5$ with condition number below 3; and `unresolved` otherwise. A readout decision spends the remaining budget without reusing pilot data.

The verdict requires three frozen criteria. H1: at least 10 decided cells, a reliable rate of at least 80%, and a Wilson lower bound of at least 50%. H2: at least 10 no-safe cells, with the Wilson lower bound of the share that avoided a failure above 50%, using full-budget raw as the fallback. H3: on decided cells, mean regret below full-budget always-raw and always-readout, with no more catastrophic cells than either.

## 2.8 Sequential certification

Two candidates with exact expectations $b \pm m/2$ are sampled in batches of 256 target shots until an interval for their gap excludes zero or the budget ends, leaving the decision unresolved. Every sequential rule tests at $\alpha/K$ with $\alpha = 0.05$ and $K$ available looks. Calibration and folded shots are charged to the same budget. We compare one final test with Wilson-based, Hoeffding gap-confidence, and empirical-Bernstein variance-aware stopping, and keep uncorrected Wald peeking only as a negative control. Readout intervals use a delta-method linearization that includes calibration uncertainty, and combining Wilson components is approximate, so validity is also measured directly at zero margin.

## 2.9 Calibration reuse under drift

A one-qubit threshold decision has true one-state probability 0.53 and threshold 0.50. Each decision spends 1,024 target shots and each calibration 1,024 shots. Assignment errors drift by a slow monotonic, abrupt, asymmetric, or random-bounded synthetic model. The first three deliberately push the corrected estimate toward the threshold, so they are stress tests; random-bounded drift is the bidirectional control. Policies recalibrate at every decision, every 10 decisions, when an age-based uncertainty bound reaches its frozen trigger, or never, and they cannot see the true drift or decision outcome.

## 2.10 MaxCut/QAOA application decision

For a graph $G = (V, E)$ and bitstring $z$, the exactly checkable cut value is

$$C_G(z) = \sum_{(u,v)\in E} \mathbf{1}[z_u \neq z_v]. \tag{5}$$

Exhaustive enumeration fixes the classical optimum and all degenerate optimal bitstrings for one width-4 ring, one deterministic width-6 sparse graph, and one deterministic width-8 medium/dense graph. A frozen grid supplies five $p = 1$ QAOA candidates per graph. Noiseless statevector expectations determine the incumbent, challengers, ideal ranks, and clear versus near-tie regimes before noisy data exist. One computational-basis shot yields one complete cut sample; edges do not receive independent free shots.

Each method receives one application-level budget $B \in \{4096, 16384\}$ for selecting among all five candidates. Raw estimation divides $B$ among candidates. Readout charges calibration within $B$, and ZNE charges all three folded executions within $B$. We report exact top-1 accuracy, top-$k$ recall, simple regret against ideal candidate energy, energy RMSE, and unresolved probability separately. At nested shot checkpoints, a candidate is certified only if its adjusted uncertainty interval is separated from every challenger under a Bonferroni simultaneous-coverage rule; the runtime selector never receives the ideal winner.

## 2.11 Hardware readiness review

For each of six candidate cells we freeze simulator predictions for all four methods, the expected decision, validity status, and uncertainty, along with a hypothesis and the outcome that would falsify it. Required shots use a two-sided normal approximation at $\alpha = 0.05$ and 80% power. The review applies seven go requirements and six no-go triggers fixed in the study plan. Every prediction is conditional on a named simulator noise condition, because a device's noise cannot be set to match it.

# 3 Experiment Setup

The main grid crosses widths $2, 4, 6$, hardware-efficient depths $1, 3, 6$, noise levels zero/low/medium/high, shot budgets 1,024/4,096/16,384, three methods, and 50 repeats. GHZ depths 3 and 6 remain explicit `not_applicable` cells. All seeds derive from root seed 20261102. Qiskit Aer supplies ideal statevectors and finite-shot noisy simulation [8]. The main model combines independent assignment error with one- and two-qubit depolarizing channels. The method comparison follows the broader need for common QEM benchmarks [2]; the scalable-readout track is motivated by assignment-matrix growth and reduced calibration models [7].

A separate 4,096-shot shift set tests intermediate mixed noise, asymmetric readout, coherent over-rotation, and correlated $ZZ$ error. These are controlled synthetic stress tests, not calibrated device models. Every manifest row has a raw row with status `complete`, `failed`, or `not_applicable`; no failed cell is silently dropped.

The combined study preregisters six diagnostic cells: readout-dominated, gate-noise-dominated, practical-tie, ZNE-invalid, 0.02-margin decision, and 0.20-margin decision conditions. It crosses budgets 4,096 and 16,384, both frozen allocation policies, four factorial methods, and 50 paired repeats. Matched-cost garbage folding is used only by the validity gate and is excluded from method ranking.

The application manifest contains three graphs, five candidates, three methods, two budgets, and 50 deterministic repeats, for 4,500 candidate-level rows and 900 selection groups. Candidate construction uses graph structure and ideal statevector values only. The exact classical solver is used for post hoc certification, never to guide the noisy selector.

The sequential study uses 13 task–margin pairs, up to four noise levels, budgets of 1,024, 4,096, and 16,384 shots, three methods, five stopping rules, and 100 repeats, for 198,000 rows. A symmetric task has equal circuit depth; the imbalanced tasks give candidates depths 6 and 1, or four times the readout error on one candidate, so bias can reverse the noisy ordering. Shots are sampled from exact channel probabilities, checked against an Aer density-matrix simulation. Alpha, batch size, and grids were written before the full grid was generated, and no threshold was tuned. The calibration study has workloads of 1, 5, 20, and 100 decisions, four drift models, four policies, and 30 repeats, for 60,480 decision rows.

The selector test uses a fresh grid committed before it ran: GHZ and hardware-efficient circuits at widths 3 and 5, depths 2 and 4 with two independent replicates, six noise conditions, budgets of 2,048 and 8,192 shots, four policies, and 50 repeats, for 120 cells and 24,000 rows.

Table 1: After correcting the oracle, raw wins most simulated cells, mainly where the reference is near zero; no method wins everywhere.

| Cost-aware label | Cells |
|---|---|
| Practical tie | 26 |
| Raw | 77 |
| Readout | 18 |
| Linear ZNE | 14 |
| Inconclusive | 9 |

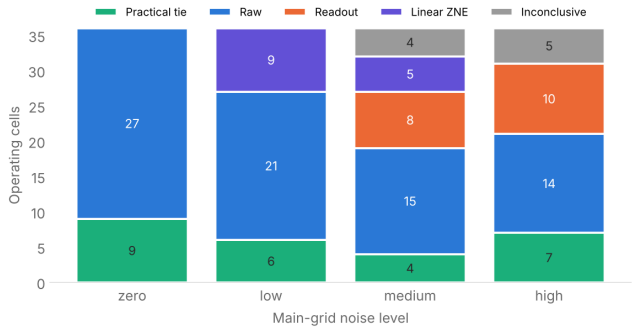



Figure 2: Corrected labels show raw winning most noisy hardware-efficient cells, while readout wins strong-signal GHZ cells at medium and high noise.

# 4 Results

## 4.1 Operating regimes

Table 1 shows 77 raw wins, 26 true practical ties, and distinct readout and ZNE subsets. Nine cells are inconclusive, eight because the 50-repeat completeness contract is not met. Raw wins require care: 40 win on accuracy and 37 on cost alone, and 63 have $|\mu_{\text{ref}}| < 0.06$, where noise moves the estimate toward a reference already near zero. They do not show that raw recovers a strong signal. Figure 2 shows how the labels vary with simulated noise.

Mitigation can also worsen estimation: 179 of 432 method summaries have negative RMSE gain relative to raw estimation. Readout produces 99 explicit hardware-efficient failures, concentrated at width 6, 1,024 shots, and medium/high noise. These failures are part of method performance, not missing data.

## 4.2 ZNE validity under negative controls

The probability-level diagnostic contains 800 simulated repeat records across 16 preregistered cells. Of these, 197 repeats are classified as extrapolation-unstable and 5 as signal-destroyed. The cell-level validity gate accepts 5 cells and rejects 11. All 8 hardware-efficient cells are rejected; their apparent corrections do not survive the combined gain-uncertainty and trend requirements. The

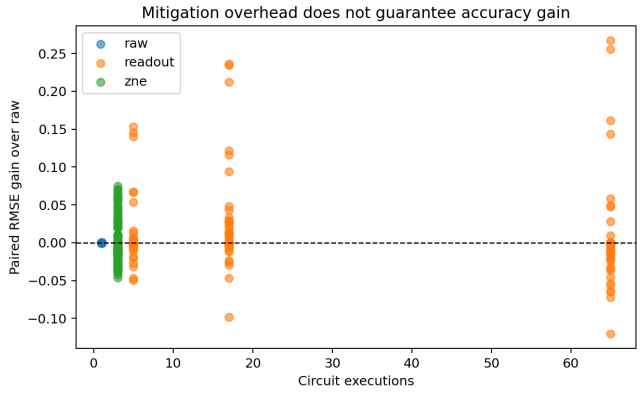



Figure 3: Higher simulated execution overhead does not guarantee positive RMSE gain; negative gains remain visible rather than being clipped or filtered.

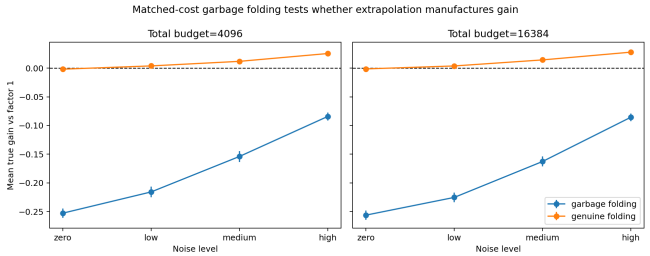



Figure 4: Matched-cost garbage folding separates an apparent correction from a unitary-equivalent ZNE effect; only cells passing the full preregistered gate are reported as credible.

five accepted cells are Bell-state conditions, so this result narrows rather than broadens the evidence for linear ZNE.

## 4.3 Combined QEM under one budget

Combined mitigation has the lowest expectation RMSE in 12 of 16 preregistered circuit–budget–allocation conditions. This descriptive win count is insufficient for a mechanism claim: combined RMSE gain versus raw ranges from negative to positive, and additional execution burden does not guarantee improvement (Fig. 6).

Exactly 12 of 24 paired factorial contrasts are negative; the remainder are positive. Their uncertainty intervals frequently overlap zero (Fig. 7), so the data do not support a uniform cooperative or antagonistic interaction. More decisively, all 16 of 16 combined-ZNE conditions are rejected by at least one frozen signal, physicality, uncertainty, trend, or garbage-control requirement. The RMSE improvements therefore remain simulated estimator outcomes rather than validated ZNE effects.

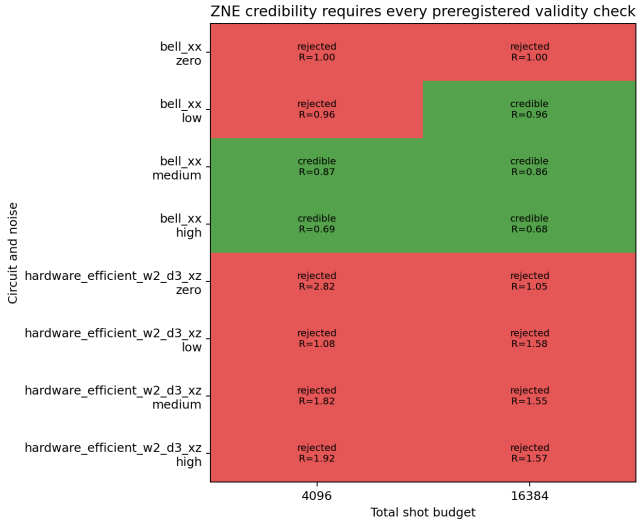



Figure 5: Signal retention and extrapolation stability vary by simulated circuit, noise, and budget; the hardware-efficient cells fail the combined validity gate despite some favorable individual diagnostics.

Table 2: Even without charging its diagnostics, the selector has more harmful outcomes than always-raw across all 180 evaluated cells.

| Category | Sel. | Raw | Read. | ZNE |
|---|---|---|---|---|
| Correct, useful | 63 | 106 | 19 | 18 |
| Correct, tied | 20 | 23 | 11 | 24 |
| Incorrect, negligible | 29 | 0 | 71 | 3 |
| Incorrect, material | 42 | 26 | 45 | 78 |
| Catastrophic | 17 | 16 | 25 | 48 |
| Abstained, avoided failure | 1 | 0 | 0 | 0 |
| Abstained, unnecessary | 5 | 0 | 0 | 0 |
| Inconclusive | 3 | 9 | 9 | 9 |

## 4.4 Selector consequences

On 29 held-out cells, the selector reaches 41.4% decision compatibility and 31.0% strict label agreement; under the four noise shifts these are 50.0% and 47.2%. Table 2 and Fig. 8 show what those choices cost.

Averages hide the worst errors. Holdout regret has median 0.0004 and mean 0.0210, yet reaches 0.2555. Of 17 catastrophic cells, 8 are high-noise GHZ cells where the selector chose raw and readout was far better. The other 9 meet only the relative rule, at references near zero. Mean holdout regret is 0.0077 for always-readout, 0.0210 for the selector, and 0.0219 for always-raw. Under noise shift it is 0.0051 for always-raw, 0.0081 for the selector, and 0.0101 for always-readout.

Across all cells, the selector has 59 material or catastrophic outcomes and always-raw has 42. That does not make raw a general policy: all of raw's harmful outcomes occur in the 84 cells with $|\mu_{\text{ref}}| \geq 0.06$, where the counts

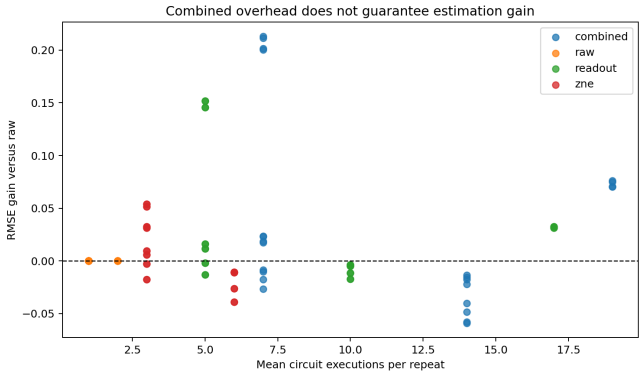



Figure 6: Combined readout correction and ZNE often lowers simulated expectation RMSE, but some conditions remain worse than raw despite greater execution overhead.

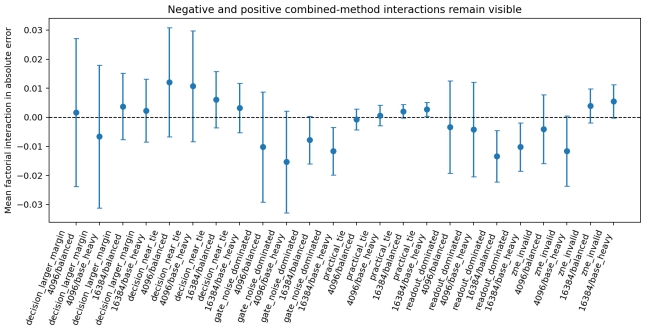



Figure 7: Paired factorial interactions change sign across preregistered cells, budgets, and allocations; many intervals cross zero, rejecting a universal synergy claim.

are 42 for raw, 28 for the selector, and 19 for always-readout. This split was chosen after seeing the data and is descriptive. Abstention remains untested in practice: the selector abstains in only 6 cells, all at width 6 and 1,024 shots, and 5 of those were unnecessary by the frozen counterfactual.

## 4.5 Reliability ranges

Table 3 lists every supported level; the last column is the share of resamples in which the level stays supported, and values below 90% are unstable. No method has a supported level on every feature, and no depth level is supported, so the frozen joint envelope is empty. A rule requiring support on every feature would always report no safe method on this grid. Margin rows come from the separate width-2, depth-3 decision benchmark.

Among diagnostics, condition number acts only as a warning: readout is reliable in 75% of cells below 1.5, 50% from 1.5 to 3, 10% from 3 to 8, and none of the 4 cells at 8 or above. ZNE fit residual does not discriminate, with rates of 24–33% in every adequately sampled bin, and the selector's residual limits are never reached. Three cells

Table 3: Only a few single-feature levels are supported, and none exists for ZNE.

| Method | Level | Reliable | Stable |
|---|---|---|---|
| Raw | width 6 | 36/44 | 55% |
| Raw | noise zero | 32/32 | 100% |
| Raw | $\lvert\text{ref}\rvert < 0.06$ | 56/56 | 100% |
| Raw | margin 0.2 | 11/12 | 93% |
| Readout | noise zero | 29/32 | 89% |
| Readout | $\lvert\text{ref}\rvert \geq 0.5$ | 26/30 | 88% |
| Readout | margin 0.2 | 11/12 | 94% |

Table 4: On fresh circuits, always-readout has the lowest mean regret and fewest catastrophic cells, while always-raw is reliable in more cells.

| Policy | Regret | Reliable | Catastrophic |
|---|---|---|---|
| Always raw | 0.0243 | 70 | 18 |
| Always readout | 0.0093 | 58 | 5 |

flip across the condition-number gate of 8 under repeat resampling, with means of 7.87, 8.10, and 8.16.

## 4.6 Preregistered selector test

All three criteria fail, so the runtime selector is a no-go. For H1, 14 of 26 decided cells are reliable (53.8%, lower bound 35.5%). For H2, 27 of 94 no-safe cells avoid a failure (28.7%, lower bound 20.6%). For H3, mean regret on decided cells is 0.0059, against 0.0606 for always-raw and 0.0033 for always-readout. The selector has 0 catastrophic decided cells, as does always-readout, and always-raw has 10; the failure is that the charged pilot leaves it behind full-budget readout.

The selector chooses readout in 22% of repeats, returns `request_more_shots` in 50%, and `unresolved` in 26%. Table 4 gives the fixed-policy comparison across all fresh cells. As an exploratory reading, raw is reliable in all 36 cells with $|\mu_{\text{ref}}| < 0.06$ and readout in 34 of 48 with $|\mu_{\text{ref}}| \geq 0.5$, but a runtime policy does not know the reference.

## 4.7 Decision reliability

Expectation RMSE and decision reliability are distinct. Figure 9 reports sign-decision error with Wilson uncertainty. A method may reduce continuous estimation error without reducing the probability of a wrong downstream decision.

The controlled combined-QEM decision study sharpens this separation. At margin 0.02, combined decision-error probability ranges from 0.26 to 0.48 across budgets and allocations and does not consistently beat readout alone. At margin 0.20, nearly all methods resolve the decision.

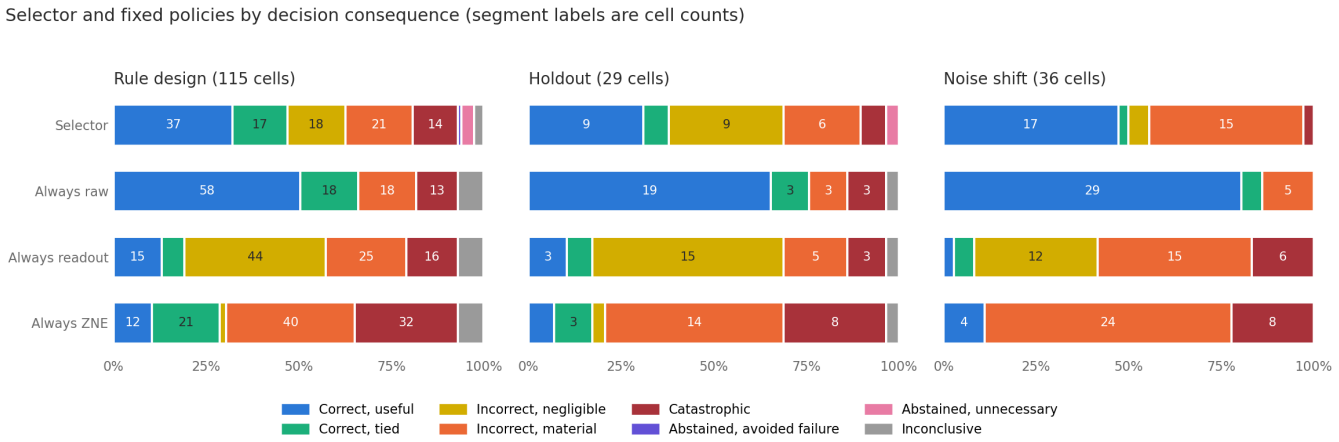



Figure 8: Label agreement hides the size of errors: the selector still makes material and catastrophic errors, and no policy dominates across every evaluation set.

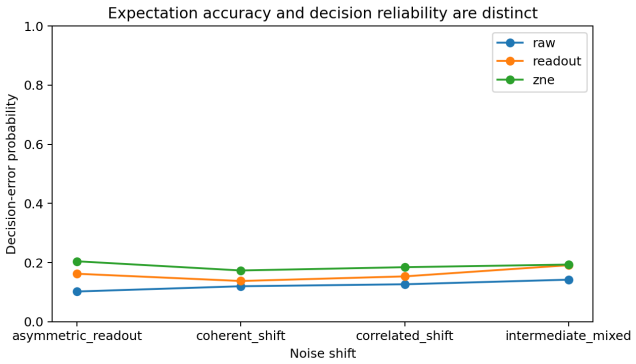



Figure 9: Decision-error probability changes by method and synthetic noise shift, showing that expectation accuracy is not a sufficient proxy for decision reliability.

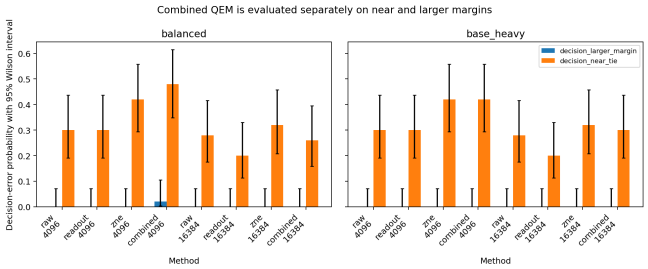



Figure 10: The larger decision margin is resolved, whereas the 0.02 near-tie retains substantial Wilson-bounded error and no consistent combined-QEM advantage.

Combined mitigation can therefore improve an expectation estimate while leaving a resource-constrained near-tie unresolved.

## 4.8 Scalable readout correction under model mismatch

The scalable-readout track compares full assignment, tensor-product, observed-subspace, and matrix-free observed-subspace correction. Its final stress test contains 9600 paired simulated rows. All methods share target counts and one fixed total budget; the calibration model remains independent while the target channel is matched independent, asymmetric, pair-correlated, or deliberately mismatched. Bias and repeat sampling spread are reported separately.

In the representative width-6 broad-output, 16,384-shot cell, deliberate calibration mismatch raises the 0.05-margin threshold-decision error to 0.88–0.94 across methods. The matched independent condition remains near zero. Figure 11 therefore shows that classical memory reduction and solver convergence do not establish readout-model validity. Decision risk is evaluated separately from expectation RMSE, consistent with recent cost-aware QEM analysis [9].

## 4.9 MaxCut/QAOA candidate selection

All completed clear-ranking selections identify the ideal best candidate. Near ties separate estimator quality from decision quality (Fig. 12). At 4,096 shots, near-tie error is 0.28 for raw, 0.32 for readout among completed selections, and 0.36 for ZNE; at 16,384 shots it is 0.17, 0.179, and 0.42. On the completed 4,096-shot ring and sparse strata, readout lowers energy RMSE from 0.183 to 0.148 and from 0.228 to 0.191, respectively, yet worsens top-1 selection on both; its medium/dense stratum fails entirely. Across completed repeats, the RMSE–selection-error correlation is only 0.153. Better energy estimation therefore does not imply a better discrete choice in this study.

Simple regret is smaller when the wrong candidate is nearly equivalent, so it and exact top-1 accuracy lead to different emphasis (Fig. 13). At 16,384 shots, mean regret is 0.0130 for raw, 0.0125 for completed readout selections,

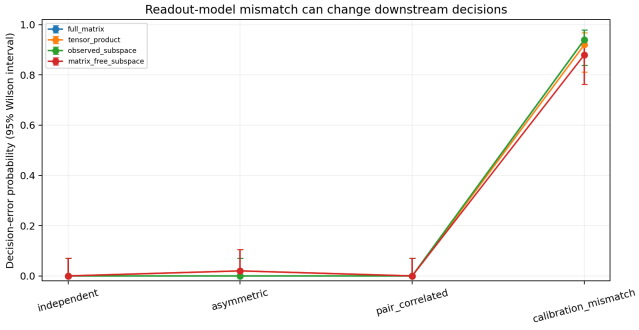



Figure 11: A deliberately mismatched simulated calibration channel causes high 0.05-margin decision error for every correction strategy; reduced storage does not repair model misspecification.

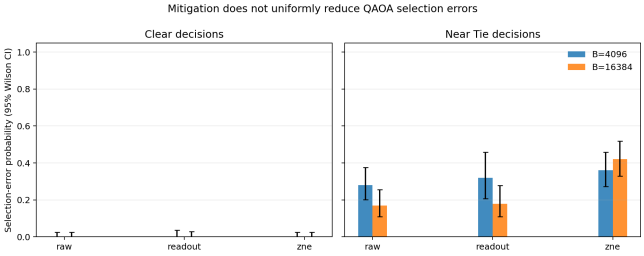



Figure 12: Clear QAOA candidate rankings are selected correctly, whereas near-tie error remains substantial and ZNE worsens when the larger budget is used.

and 0.0378 for ZNE. Readout fails all 50 medium/dense selections at 4,096 shots and 22 of 50 at 16,384 shots because its assignment correction is ill-conditioned; these 72 failed groups are not treated as abstentions or successes. Results also vary by instance: ZNE's 16,384-shot top-1 error is 0.08 on the ring, 0.34 on the sparse graph, and 0.50 on the medium/dense graph.

Certification is much stricter than selecting a point-estimate winner. Only two of 828 completed selection groups certify before exhausting the declared budget, both raw width-4 ring cases at 16,384 shots. The remaining 826 completed groups are unresolved, and 72 groups fail explicitly (Fig. 14). QEM overhead therefore buys no scientifically meaningful certification improvement in this frozen application grid.

## 4.10 Sequential certification

Corrected looks control false certification. At margin 0, where any certification is false, Wilson stopping certifies in 1.7%–2.7% of repeats across methods, whereas uncorrected peeking reaches 22.4%–24.9%. These pooled zero-margin cells use shared calibration, so per-candidate readout intervals have no direct null test. Table 5 shows the trade-off on the symmetric task: early stopping saves shots, yet leaves more decisions unresolved than a single

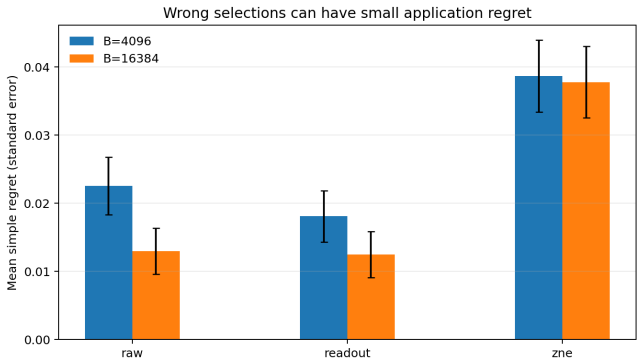



Figure 13: Application regret and exact top-1 error are distinct: a wrong near-tie choice can have low regret, while mitigation overhead does not ensure lower regret.

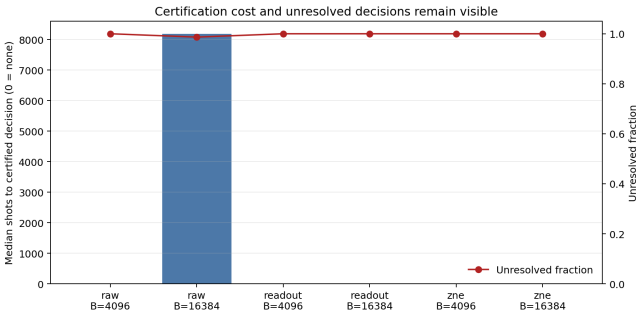



Figure 14: Simultaneous-interval certification is achieved in only two completed selections; nearly every completed method–instance–budget repeat remains unresolved.

full-budget test because each look is stricter. Mitigation delays certification when the estimator already has the right sign. With shared calibration, readout correction cannot change the sign of the gap, so it only removes target shots.

Near ties consume the budget (Fig. 15): at margin 0.02, Wilson stopping uses 98.1%–99.8% of 16,384 shots and leaves 96.0%–99.7% of repeats unresolved. Certification also does not detect bias (Fig. 16). With strong gate-noise imbalance at margin 0.05, raw certifies the wrong candidate in 100% of repeats after a mean of 2,022 shots, and ZNE still does so in 77%. Readout correction removes this reversal when the imbalance is in readout error, at the cost of more unresolved decisions. A certificate therefore says on which side of zero the estimator's own limit probably lies, not that the exact winner has been found.

## 4.11 Calibration reuse under drift

Table 6 compares policies over 100 decisions, averaged across drift models. Triggered recalibration has decision error close to recalibrating every time at about 40% lower cost. Never recalibrating is cheapest but least reliable, and that depends strongly on drift direction (Fig. 17): its

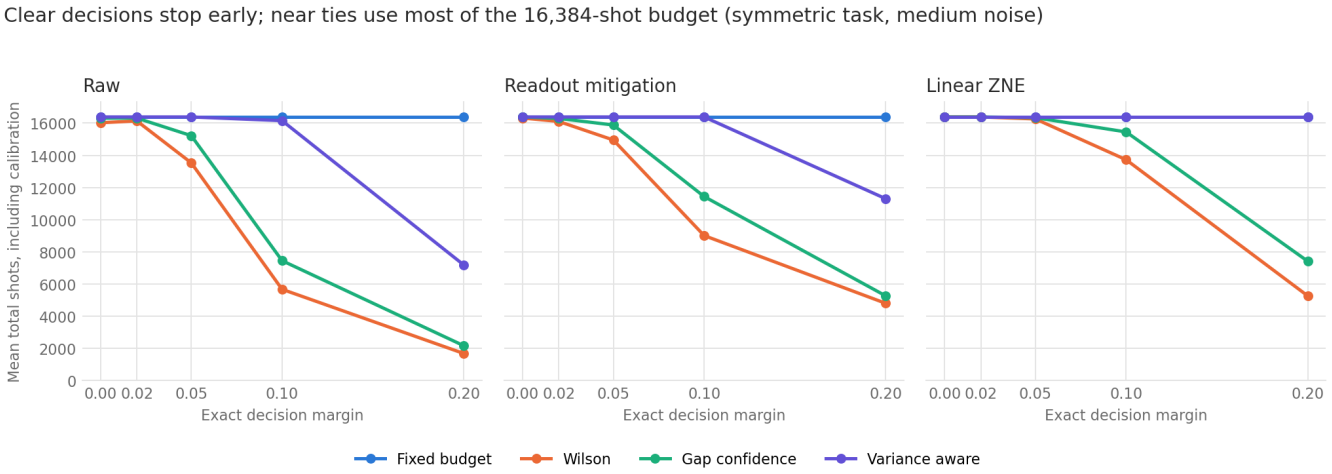



Figure 15: Clear decisions stop early under corrected rules, whereas near ties use nearly the whole budget; mitigation shifts every curve toward more shots.

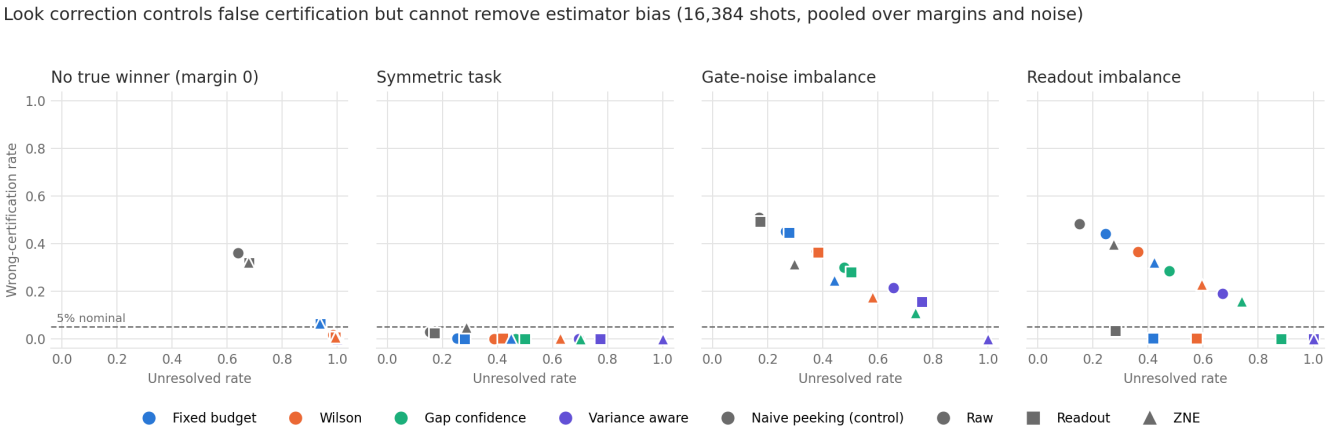



Figure 16: Corrected looks keep false certification low when no winner exists, but stopping rules cannot remove estimator bias, as the imbalanced tasks show.

error is 0.386 under abrupt change, 0.331 under asymmetric drift, 0.170 under slow drift, and 0.047 under the bidirectional control.

## 4.12 Hardware readiness

Table 7 lists the frozen predictions. The review decision is no-go: only 1 of 7 go requirements is fully met, and all 6 no-go triggers apply. The plan would need about 72.1 million shots, with no budget declared. The near-tie cell alone needs 205,218 shots per repeat and is already inconclusive at 16,384. The positive control is credible at 16,384 shots but rejected at 4,096, three of five ZNE-relevant cells have no single-method validity assessment, and no code captures device metadata. This is a measured outcome of the review, not a statement about how any device would behave, since no job was submitted.

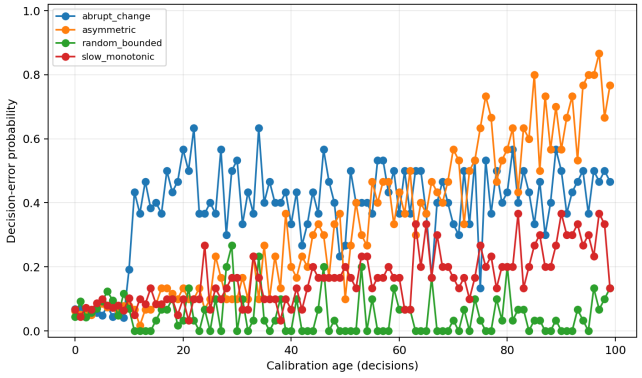



Figure 17: Decision error grows with calibration age under directed synthetic drift but not under the bidirectional control.

Table 5: Stopping early saves shots but leaves more symmetric-task decisions unresolved; mitigation delays certification further (16,384 shots, margins above 0, nonzero noise).

| Method | Policy | Shots | Unres. | Wrong |
|---|---|---|---|---|
| Raw | Fixed budget | 16,384 | 27.1% | 0.08% |
| Raw | Wilson | 9,583 | 40.7% | 0.00% |
| Raw | Gap confidence | 10,653 | 48.2% | 0.00% |
| Raw | Variance aware | 14,136 | 70.8% | 0.00% |
| Raw | Naive peeking | 5,067 | 16.2% | 3.25% |
| Readout | Wilson | 11,429 | 43.7% | 0.08% |
| ZNE | Wilson | 13,045 | 63.6% | 0.08% |

Table 6: A triggered policy approaches every-decision reliability at lower cost, while indefinite reuse is unsafe on average under these stress models (100 decisions).

| Recalibration | Shots per decision | Error |
|---|---|---|
| Every decision | 2,048 | 0.062 |
| Every 10 decisions | 1,126 | 0.074 |
| Uncertainty trigger | 1,229 | 0.067 |
| Never | 1,034 | 0.234 |

### 4.13 Evidence by track

Table 8 separates each finding by evidence type. The hardware column is empty for every track.

## 5 Conclusion

No QEM method is uniformly useful under the declared fixed-budget simulator grid. Raw wins most cells, largely where the reference is near zero, some cells favor readout or ZNE, and low-shot readout calibration can fail explicitly. The frozen rule selector is reproducible and explainable, but it does not clearly beat fixed policies even without being charged for its diagnostics, so it is a baseline for further study rather than a successful adaptive policy. A preregistered runtime selector also fails on unseen circuits, so simple fixed policies remain the reference that any adaptive rule must beat. The strongest supported conclusion is operational: mitigation should be compared with raw estimation under equal resource accounting, with ties, negative gains, abstention, and failed ZNE validity checks preserved. In this benchmark, apparent extrapolation gains are more common than gains that survive the full falsification gate. Combining readout correction with ZNE improves RMSE in selected simulated conditions, but neither its factorial interaction nor its downstream decision error supports a universal synergy claim, and no combined condition passes the frozen validity gate. The MaxCut application strengthens that boundary: lower energy RMSE is not a reliable proxy for top-1 selection,

Table 7: Frozen simulator predictions for proposed cells at 16,384 shots, or a 100-decision workload for calibration reuse. These are hypotheses for a device, not hardware results. Val. is single-method ZNE validity; Mshots is the planned total in millions.

| Cell | Raw | RO | ZNE | Comb. | Val. | Mshots |
|---|---|---|---|---|---|---|
| Readout-dominated | 0.740 | 0.893 | 0.791 | 0.958 | credible | 3.3 |
| ZNE control | 0.970 | 0.987 | 0.980 | 0.997 | credible | 4.1 |
| ZNE signal risk | 0.232 | 0.263 | 0.265 | 0.305 | none | 4.1 |
| Near tie | 0.012 | 0.021 | 0.014 | 0.021 | none | 41.0 |
| Clear margin | 0.179 | 0.181 | 0.184 | 0.188 | none | 3.3 |
| Calibration reuse | 0.510 | 0.530 | – | – | – | 16.3 |

simple regret, or certified decision, and the added QEM executions do not produce a general application advantage. Sequential stopping can save shots with corrected looks, but a certified decision is only as sound as the estimator's bias, and calibration reuse is safe only for the drift it was tested against. A device run is not yet justified: without a declared budget, metadata capture, and complete validity controls it could not separate method failure from device drift.

## 6 Discussion

### 6.1 Supported findings

The data directly support regime dependence, nontrivial execution overhead, explicit readout instability, negative mitigation gains, and a gap between estimation RMSE and decision reliability. The deterministic holdout also shows that intuitive rules do not generalize reliably across the present circuit and noise grid. The ZNE negative control further shows that improvement toward a reference is insufficient evidence unless signal retention, physicality, uncertainty, and a matched-cost control agree, matching concerns about artefactual ZNE improvement [5]. The combined experiment further supports a separation between estimator performance and mechanism validity: lower RMSE can coexist with rejected ZNE diagnostics and unresolved near-tie decisions.

### 6.2 Hypotheses and limitations

Commit order proves preregistration only for the combined, QAOA, scalable-readout, and runtime-selector studies. The ZNE gate's checks were named before its results, but its numeric thresholds first appear with them, and the sequential and calibration rules also first appear with their results. The selector's freeze before holdout evaluation is likewise a process claim: its rules and first results appeared in one commit, so version history alone cannot prove the order. We found no evidence of tuning, but the corrected analysis is not an independent test. The later runtime-selector test has that independence for its

Table 8: Final evidence matrix. Y marks evidence of that type; An. is analytic, NL noiseless simulation, NS noisy simulation, HW hardware. Findings are measured in simulation, and unresolved questions are hypotheses.

| Track | An. | NL | NS | HW | Supported finding | Unresolved question | Limitation |
|---|---|---|---|---|---|---|---|
| Expectation accuracy | Y | Y | Y | – | No universal winner: 77 raw, 26 ties, 18 readout, 14 ZNE. | Which method wins when the signal is strong and the circuit is new? | Raw wins are mostly near-zero references; one circuit per width and depth. |
| Decision reliability | Y | Y | Y | – | Lower RMSE does not ensure a better decision; near-tie error is 0.26-0.48 for combined QEM. | How does this extend beyond two candidates? | Controlled margins on small circuits. |
| ZNE validity | Y | Y | Y | – | Apparent gain is not validated gain: 5 of 16 cells pass, none hardware-efficient. | Does the gate hold for other circuits and fits? | Two circuits, linear fit, and folds 1, 3, and 5. |
| Combined mitigation | – | Y | Y | – | Lowest RMSE in 12 of 16 conditions, yet all 16 validity checks fail. | Are readout and gate errors separable on a device? | Six cells, two budgets, two allocations, higher overhead. |
| Readout scalability | Y | Y | Y | – | Lower memory is not robustness: mismatch error reaches 0.88-0.94. | How large is real correlated readout error? | Synthetic channels; correction amplifies sampling spread. |
| Application case study | Y | Y | Y | – | Lower energy RMSE does not imply better selection; 2 of 828 completed selections certify. | Does this hold for other graphs and depths? | Three graphs, five frozen candidates, and p=1. |
| Calibration drift | – | – | Y | – | Triggered recalibration saves about 40% of shots with similar error; never recalibrating reaches 0.386 under abrupt drift. | What drift does a real device show? | Directed synthetic stress; not a physical lifetime. |
| Sequential certification | Y | Y | Y | – | Corrected looks keep false certification at or below 2.7%, versus up to 24.9% for peeking; bias is not detected. | How can bias be detected before certifying? | Two-qubit circuits, one observable, conservative bounds. |
| Hardware readiness | – | – | Y | – | Review decision: no-go. No job was submitted. | Can a backend-specific, powered, controlled protocol be justified? | Predictions depend on simulator noise; no backend data. |

own rules, but it covers one design, and its H2 fallback may be too lenient for sign decisions near zero, a hypothesis that needs a new preregistration. Possible causes of selector error include insufficient features, first-order ZNE, and the mismatch between synthetic training and shift channels; the experiment does not isolate these causes. The validity gate is conditional on its frozen thresholds, linear fit, chosen folds, and maximally mixed observable-floor model; it does not establish a general theorem about ZNE. All noisy values are simulator-derived. There is no IBM or other hardware execution, device drift, calibrated crosstalk, leakage, relaxation, or non-Markovian noise. Width is limited to six, the hardware-efficient ansatz and observables are fixed, and full assignment calibration scales exponentially. The same seed family and simulator implementation bound generalization. No ML selector is trained. The hardware-efficient grid has one circuit per width and depth, so width and depth effects are confounded with specific angles, and reliability ranges are marginal values on a coarse grid that must not be interpolated or extrapolated. The combined study covers only six preregistered cells, two budgets, and two allocation policies; it does not compare extrapolate-then-correct ordering or establish that readout and gate errors are separable on hardware. The scalable-readout mismatch study is limited to synthetic disjoint pair flips, one independent calibration approximation, widths at most six, and a single fixed decision margin; its negative result cannot quantify hardware crosstalk or drift. The application study covers only three deliberately controlled graph instances, five ideal-only candidates per instance, and $p = 1$. This curation is useful for known-reference falsification but is too small to support graph-family or scaling claims. Hardware evidence would require frozen candidates and budgets followed by device jobs that record calibration time, topology, transpilation, drift, leakage, and crosstalk; none of those effects is measured here. The QAOA candidates are also frozen inputs: exact energy ties are ordered by floating-point noise from the machine that generated them, so the selected candidates are not derivable from scratch on another platform, although no noisy outcome influenced them. The sequential study covers two-qubit circuits and one observable, and its bounds use a union over looks rather than an optimized spending rule. The calibration study is a one-qubit threshold task under directed synthetic stress models; it does not establish a physical calibration lifetime.

## 6.3 Distinctions kept separate

Expectation RMSE and decision error are reported separately because one can improve while the other does not. Apparent ZNE gain counts only movement toward a reference, whereas validated gain must also pass the frozen gate. Single-method and combined-method costs are charged in the same ledger: at width 2 and 16,384 shots, combined mitigation uses 7 circuit executions compared with 1 for raw, 3 for ZNE, and 5 for readout. Measured values come from committed rows; the causes offered above and the predictions in Table 7 remain hypotheses. All evi-

dence is analytic or simulated, so nothing here is hardware evidence.

### 6.4 Why the review ends in no-go

A disagreement between a device and Table 7 would first falsify the transfer of a synthetic noise model, not the simulated conclusion itself. Separating drift from method failure needs interleaved circuits, repeated reference controls, and a calibration record around each block, none of which exists here. One backend could support at most a time-bound claim about that device. Separate jobs can cross a calibration update, so paired comparison would hold only within blocks that do not. The power calculation is also optimistic for a device because its effect sizes are simulated, and the drift cell cannot be powered without a drift estimate. A run now would mainly produce a demonstration, so a no-go is the more honest result. Reopening the review requires a backend-specific preregistration, declared budget, and working metadata capture.

### 6.5 Manifest deviation

The scalable-readout manifest planned 960 repeats of correlated readout at widths 8–12 that were never run; correlated models ran only at widths 2–6, in the separate mismatch study. These rows are reported as not executed.

## References


[1] S. Bravyi, S. Sheldon, A. Kandala, D. C. Mckay, and J. M. Gambetta. Mitigating measurement errors in multiqubit experiments. *Physical Review A*, 103(4):042605, 2021.

[2] D. Bultrini et al. Unifying and benchmarking state-of-the-art quantum error mitigation techniques, 2021.

[3] Z. Cai, R. Babbush, S. C. Benjamin, S. Endo, W. J. Huggins, Y. Li, J. R. McClean, and T. E. O'Brien. Quantum error mitigation. *Reviews of Modern Physics*, 95(4):045005, 2023.

[4] T. Giurgica-Tiron, Y. Hindy, R. LaRose, A. Mari, and W. J. Zeng. Digital zero noise extrapolation for quantum error mitigation. In *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316, 2020.

[5] D. Köster and W. Mauerer. Benchmarking error mitigation: Artefactual improvements in zero-noise extrapolation, 2026.

[6] Y. Li and S. C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Physical Review X*, 7(2):021050, 2017.

[7] P. D. Nation, H. Kang, N. Sundaresan, and J. M. Gambetta. Scalable mitigation of measurement errors on quantum computers, 2021.

[8] Qiskit Aer contributors. Qiskit aer: High-performance quantum circuit simulators. https://qiskit.github.io/qiskit-aer/. Software documentation, accessed 2026-11-02.

[9] V. Scavino Alfaro. Accuracy, decision risk, and sampling cost in quantum error mitigation, 2026.

[10] K. Temme, S. Bravyi, and J. M. Gambetta. Error mitigation for short-depth quantum circuits. *Physical Review Letters*, 119(18):180509, 2017.


## A Reproducibility

```
python -m pytest
python -m src.verify_generated_artifacts
python -m src.verify_generated_artifacts --phase-f
python -m src.sequential_decision
python -m src.calibration_drift
python -m src.regime_labels_v2
python -m src.selector_audit
python -m src.reliability_envelopes
python -m src.generate_paper_assets
python -m src.generate_selector_holdout_v2
python -m src.generate_audit_paper_assets
python -m src.hardware_readiness
python -m src.final_synthesis
tectonic paper/main.tex --outdir paper
python -m src.verify_final_project
python -m src.claim_ledger
```

Repeat-level rows remain under `results/raw`; processed summaries, manifests, run metadata, seeds, diagnostics, and execution counts remain separate and auditable. Tests check analytic references, statevector agreement, folding equivalence, shot accounting, deterministic seeds, manifest coverage, metric recomputation, and the holdout boundary. `src.verify_final_project` recomputes the main summaries from raw rows at tolerance $10^{-9}$, accounts for every manifest, and checks figures and this document. `src.claim_ledger` separately recomputes the printed values of computed claims, shares no code with the generators, and labels review judgments, manual interpretation, hypotheses, and limitations.